Washing

Washing: It is a process or technology which is used to modify the appearance, color, outlook, comfortability, design of the garments & to remove all the impurities those are attached to garments during manufacturing as well as to remove the size materials from the garments.

Objectives of washing:

→ to remove all ~~the importance~~ impurities those are attached with garments parts during manufacturing (Spreading, cutting, sewing)
→ to remove size materials when needed.
→ to get soft hand feel.
→ for color fading effect.
→ to produce fashion apparel according to buyer's requirements.

Types of m/c used in washing plant:

① Garments washing m/c
- → Horizontal loading
- → vertical loading.
- → front loading
- → side loading.

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② Hydroextractor m/c
③ Garments Dryer : steam & Gas
④ Industrial oven : Gas / electric
⑤ pump
⑥ Boiler for steam generation.

features of pumice stone:

→ perforated stone.
→ Having irregular & rough surface
→ produced from volcanic explosion
→ floats on water
→ Having ability to soak solution due to perforation
→ These are available in 3 sizes

Small 2-3cm
Medium 3-5cm
large 5-7cm

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classification of Apparel washing :

Garments washing

wet process / Chemical process

eg: - Normal wash / Garments wash
- pigment wash
- caustic wash
- Bleach wash
- Acid wash
- Enzyme wash
- Stone wash
- Stone Bleach wash
- Stone enzyme wash
- Silicon wash
- Super white wash etc

Dry or mechanical process

eg:
- Sand blasting
- Hands scraping
- Overall wrinkling
- permanent wrinkling
- Broken & tagging
- Grinding & destroying
- PP Spray & PP Spanging
- Ozone
- laser
- whiskering

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process flow chart for Apparel Washing;

Batch / lot selection

↓

loading in washing m/c

↓

pretreatment / desizing

↓

~~Hot wash~~

&

water (M:L) selection

↓

Chemical washing

↓

~~Hot wash~~ neutralization

↓

Softening

↓

Unload

↓

Hydro extracting

↓

Drying

↓

Quality check

Normal Wash

Normal wash Normal wash is the most simple type of wash with the lowest washing cost. It is the process in which heavy or slight soiling is removed & transferred to the water in the form of a solution of dispersion.

flow process: Desizing → Rinse wash → softening → H.E → Drying → Delivery

procedure:

① first step: Desizing

Lot size: 70kg

Add water: 700 litre (M:L = 1:10)

Detergent: 1g/L

Desizing Agent: 1g/L

Temp: 40-60°c

Time: 20-25 min

Drop the liquor,

Cold wash

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2nd step: softening

Add water - 420 litre (1:6)

flax softener:- 0.6 g/L

Acetic Acid: 0.5 g/L

Time: 5 to 10 min

Drop the liqour & unload from the washing m/c.

3rd step: Hydroextracting

→ to remove the excess water from the Garments.

4th step: drying

load on steam drier

Temp: 60-70°c

Time: 40-50 min for dry

10-15 min for cold dry

5th step; Delivery

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Bleach Wash

Bleach: Bleach is a chemical that removes colors from the garments.

Common types of bleaching agent:

- Bleaching powder [$Ca(OCl)Cl$]
- H_2O_2
- $NaOCl$
- O_3

Bleach Wash: It is a one type of washing process which is done by using bleaching agent & helps to remove color from garments. Various Colors fadded effect can be get by this wash & it can be applied on all types of garments such as woven, Knit etc.

purpose of bleaching:

→ to acheive fadding effect on Garments.
→ to remove size materials from Garments
→ to acheive soft hand feel
→ to increase color fastness & rubbing fastness
→ partial feding is done on the garments

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process flow chart:

Batch selection → loading → desizing → Bleaching → neutralization → Softening → unload → Hydro extracting → Drying → QC

procedure

1st Desizing: Same as ~~be~~ desizing of Normal wash

2nd step: Bleaching

~~Or~~

Bleaching powder 10g/L
soda ash 5g/L
Temp 60°c
Time: depends on shade

Or, H_2O_2 – 12g/L
stabilizer 5g/L
Temp – 85–90°c
Time : 10–15min

3rd step: Neutralisation

Sodium hyposulphite ~~3~~ g/L (for cl bleach)
Acetic acid 1g/L for peroxide bleach

4th Softening: Same as softening of normal wash

5th: Hydroextracting

6th: Drying

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Scepero white wash

Scepero white wash is done on the garments made from grey cloth to make it extremely white.

Flowchart:

Batch selection → Loading → Desizing → Bleaching → Refine Bleaching → Neutralisation → OBA using → Softening → Unload → Hydro extracting → Drying

procedure:

1st step: Desizing : Same as Before

2nd step – Bleaching: H_2O_2 12 g/L
Stabilizer 5 g/L
Temp – 90°c
Time – 40-60 min

3rd step – Refine Bleaching:
Add H_2O_2 6 g/L
stabilizer 2 g/L
Temp 60°c
Time – 45-60 min

4th step: Neutralization: Add acetic acid 1 g/L
Time 5-10 min

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5th step: optical Brightening:

Add OBA 0.6 g/L

Time ½–10 min

6th Softening: আগের মত

7th: Hydro extracting

8th step: Drying (same as before)

Enzyme wash

Enzyme: Enzymes are living organisms or bio chemical substances that behaves like a catalyst towards specific reaction. We use it in gaurment wash to get soft hand feel & regulam fading effect.

The action of enzyme during enzyme wash, it hydrolyzes the cellulose. At first, it attacks the projecting fibers & hydrolyzed them. Then it attacks the yarn portion inside fabric & partly hydrolyzed them. As a result, Color comes out from the yarn portion & faded effect is produced

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Advantages:

→ environmental friendly treatment
→ simple process handling
→ minimum effluent problem
→ less corrosive
→ use of softener can be minimized.
→ prevent tendency of pilling.

Disadvantages: → more expensive
→ reduces strength of fabric.
→ reduces serviceability of garment
→ more time consuming process.

procedure for ~~light shad~~ Dark shade:

1st step: Desizing (আকার মণ্ড)

2nd step: Enzyme

Acid enzyme 1 g/L
Anti stain 0.6 g/L
Acetic acid 0.6 g/L
Temp → 45°c
Drop the liquor.

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~~3rd: Neutralization (Acetic acid 1g/L)~~

~~4th~~ 3rd step: Softening: (আগের মত)

5th: Hydro extracting

6th: Drying (আগের মত)

7th: Delivery]

For Medium shade

1st step: Desizing (আগের মত)

2nd step: Enzyme (আগের মত)

3rd step: Bleaching

H_2O_2 — 12g/L

stabilizer = 5g/L

Time → 15-20 mm

Temp — 50-60°C

4th step: Neutralization

Acetic acid 2g/L

Time 10-15 mm

5th: Softening

6th: Hydroextracting

7th: Delivery

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For light shade:

1st: Desizing (same)

2nd: Enzyme (same)

3rd: Bleaching (same)

4th: Neutralization (same)

5th: Hot wash

Temp 60°C

Add H_2O_2 4g/L

Caustic soda 1g/L

Time 10min

6th: softening (same)

7th: Hydroextracting

8th: Drying (same)

9th: Delivery.

Shortcut:

Dark shade → just enzyme

Medium " → enzyme + bleach + neutralization

Light " → enzyme + bleach + neutralization + Hotwash

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Acid Wash

Acid wash

→ Acid wash is done by tumbling the garments with pumice stones persoaked in a solution of phosphoric acid (H_3PO_4) & potassium permanganate ($KMnO_4$) to create an irregular color fading effect

→ pumice stones are used for heavy jeans, denim canvas & twill garments.

→ Incase of knitted fabric, thermoplastic ball is used but it gives a regular color fading effect.

Recipe: water → 100L

$KMnO_4$ → 1L ($\frac{1}{100}$th of water)

H_3PO_4 → 250ml ($\frac{1}{4}$th of $KMnO_4$)

pumice stone = $\frac{1}{2}$ vol^m of Lot

Temp = 40-50°c

Time = 1-2 hours for soaking solution by the stone and 30 min for tumbling in washing m/c

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Procedure:

→ Select the lot or batch of garments.

→ Add water, $KMnO_4$ & phosphoric acid & ~~stir~~ stire it in a stainless steel tube with dry pumice stone.

→ After soaking, the pumice stones are collected & kept at air about 1-2 hours.

→ placed the stone with garments in washing m/c and run the m/c.

→ tumble for 30 min. In this stage, the solution from the pumice stone starts to come out & rubbing action takes place between the stone & garments.

→ The normal wash is done then.

→ for the neutralisation, Add sodium metabi-sulphite ~~5(g/L)~~ (5 g/L) in cold temp to neutralize the $KMnO_4$

→ softening is done by adding softener abaut 1 g/L

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→ unload the garments & passes through hydroextraction & ~~dryer~~ Drier ~~mle~~ dryer machine.
Then it is passed to the quality check dept.

process control parameters of Acid wash:

→ size of the pumice stone
→ quantity of pumice stone.
→ Time.

Stone wash (Stone bleach wash same)

Stone wash means the process with porous volcanic stone ~~is~~ which is used to give the garment a strong & rough wash to achieve the pronounced wash effect through abrasion.

Objects:

→ to create irregular fading effect
→ to remove impurities
→ to get soft hand feel.

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~~60kg~~ for 60kg

procedure:

1st step - Desizing (same as before)

2nd step: Hot wash (~~60~~

Add water → 1:9 (540L)

Temp — 60°c

Time — 5min

drop the liquor

3rd step - Bleaching: add water - 480L (1:8)

H_2O_2 — 12g/L

Stabilizer - 5g/L

pumice stone = ½ volm of garments

Temp — 90°c

Time 12-15min

Droop the liquor

4th step- Neutralization

Add water 540L

~~Sodium h~~ Acetic acid - 2g/L

Temp - 40°c

Time - 12-15min

Drop the liquor.

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5th step: Softening

Add water - 480L

Acetic acid - 0.6g/L

Cationic softener 1g/L

Time 5 min

Drop the liquor & unload the garments lot.

6th: Hydro extracting

7th: Dryer.

8th: Delivery.

Stone enzyme (Irregular fading effect)

procedure:

1st: ~~Dest~~ Desizing (আগের মত)

2nd: Hot Wash (same as stonewash)

~~3rd:~~

Third step (Enzyme):

Add water 480L

pumice stone ½ vol^on of Garments

enzyme 1·5 g/L

Acetic acid 0·6 g/L

Anti stain 0·8 g/L

Temp - 40-60°C

Time - 60 to 70 min

Drop the liquor & Rinse twice, ~~for~~ each 3 mn

Then pumice stone is brought out from washing m/c.

4th: Bleaching

~~Bleaching pow~~ H_2O_2 12 g/L

Stabilizer 5 g/L

Temp - 90°C

Time: 12 - 15 mm

5th: Neutralisation (same as stone)

6th: softening (same as stone)

7th: Hydroextracting, 8th: Drying 9th: Delivery

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Dry process

Hand Sand blasting:

Sand blasting is one of the dry processes used in Garments washing. It is a mechanical process which creates localized abrasion or color change effect on denim, jeans, canvas, Twill, corduroy Garments by spraying Al_2O_3 with help of high pressure nozzle.

procedure:

→ Al_2O_3 is used for sand blasting which looks like sand. These Al_2O_3 are very fine & very hard.

→ Blasting action is done by spraying Al_2O_3 with the help of high speed air nozzle or gun onto the specific area of garments.

→ The blowing angle must be within 10°-20°

→ for light effect, angle keeps at 11°-12°

& for heavy effect, it keeps at 17°-18°.

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→ After mechanical fading, chemical treatments are ~~feed~~ required for the buyer's requirement

Advantages:

→ Easy process
→ requires less machinery
→ variety of distressed or abraded looks are possible.
→ Any no of design could be created by special techniques.

Disadvantages:

→ Hazzardous process.
→ Sand blasting area is kept isolated from the production area due to its noise.
→ Handling of sand blasting gun is risky process.
→ Operators should use musk, helmet, gloves, earplugs.

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Hand scraping/ Hand roubbing:

→ the process is fully manual.

→ In hand scraping method, color feding effect is produced on jeans, denim, canvas. Garments by scraping the sand paper or emery paper on the specific parts of area.

→ The most important factor is to select the right no of paper according to the fabric strength & intensity needed.

→ The sand no refers to the grit size or particle diameter in Um of sand paper. It ranges from 40-600.

→ Larger the sand no. means smoother the sand paper & the roubbing effect will not be so heavy.

Composition of sand paper: Al_2O_3, SiO_2, silicon carbide, Correndum.

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Advantage: → easy process.
→ less m/c required.
→ more used than sand blasting due to it's eco-friendlyness.

Disadvantage: → As it is fully manual, the quality will not be same.

→ it requires more time as it perform in individual garments, not batchwise.

PP spray

→ PP spray is being done on denim garments on jeans to take a bright effect on sand blast area or to create uniform color fading effect on garments.

→ At first, the pp solution is prepared by mixing $KMnO_4$ with H_2O at 1:4 ratio (80% water & 20% $KMnO_4$)

→ then the sol^n is sprayed on the specific area with the help of spray gun.

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→ this area will be greyish brown when it is neutralized neutralised by metabisulphite or oxalic acid.

pp sponging / Brushing:

→ prepare $KMnO_4$ solution (ratio 1:4)
→ take the sol^n in a tray or bowl.
→ take rubber board.
→ place the rubber board inside a leg panel of jeans.
→ Now place the jeans with rubber board on a flat table & sponged pp on apparel surface.
→ then color is will be faded.
→ After this process, wet process is essential.

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Advantage:

→ Easy process

→ less machinery is used.

Disadvantage: pp spray is the most dangerous practice for the health & safety of wortcker. It is done with a spray gun that transforms chemicals into micro particle which can be absorbed by people & resultts lung problem.

Whiskering

Whiskers are one of the most important design of used look garment. The idea of whiskers is taken from worn outline & impression pattern generated by natural wearing on hips & thigh area. It creates sitting poisture effect on hips & thigh area of denim garments.

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Procedure: Mostly rubber balloons are available with different pattern is design is used.

→ Garments are mounted on balloons & filled with air to get impression of whiskers.

→ Garments is screibed over pattern carefully with emery or sand paper on engraved pattern line.

→ Whiskering is famous for high quality & cost effectiveness.

→ This operation requires high skilled workers who can handle uniformity & sequence in whisker line. fabric may damage during rubbing if proper care is not taken.

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Destroyed denim / Damages / Destruction

→ Destruction is one of the most famous dry processes which ~~creates~~ make denim look unique & different than other.

→ To make destruction pen type of stone tools being used in mid of wash process to apply an desired area.

→ It can also be done by cutting it through Knife & holes can be made by cutting warp & weft thread

→ Different types of m/c's used for destroyed garments

- Grinding m/c
- emery cloth
- Hacksaw blade
- Knife.
- Needle etc.

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Wrenkling

A wrenkle, also known as rhytide, is a fold, ridge or crease in the cloth of garments. Wrenkling is a dry process by which wrenkles are produced onto the garments for the purpose of making fashionable garments according to buyer's requirements.

There are two types of wrinkle process:

1. permanent wrenkle process
2. Temporary wrinkle process.

permanent wrinkle process:

→ Generally permanent wrinkle is done on garments after all types of wet process
→ It is done on the garments made from all types of fabric like denim, twill, canvas poplen, corduroy, knit & polyester etc.

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→ For permanent wrinkling, we use resin which is sprayed on specific area of garments by nozz

→ Resin is diluted with water which is recomm ded by chemical supplier, generally 20% resin & 80% water.

→ After resin spray on respective area, folding is done by buyer's demand & clip attatched upon the folded area.

→ Then the garments hang in to the hangers trolley & trolley with resin treated garmen are put inside the industrial oven & temp. is set at (140°-160)c for (20-40)min.

→ Start the m/c, when setting time is over, machines are automatically off. Then it is kept for 1omin for cooling. Then the garments are unloaded from oven & ~~open the clip~~ clip is openeed. & then it is delivered to the quality checking section.

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Overall wrinkling:

→ For overall wrinkling, at first ~~the wh~~ tie the whole garments in tight position by thread.

→ We used resin in the washing m/c with water & resin tied garments for 5-10 min at 50°c.

→ Then unload the garments from washing m/c to trolley for hydro extractor to remove the excess water.

→ open the tie & cut the thread.

→ ~~Now~~ the garment hang on the hanger trolley & then it is placed inside the industrial oven & temp set at 140°-160°c at 50-70min

→ Start the m/c & when setting time is over the m/c will be automatically off & kept for 10min for cooling.

→ Then unload it from oven & delivered to the quality checking section.

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